

Mathematics of Intelligence

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Preface

Who I am

I am Martin M. W., born in March 1996. I studied software engineering at Nanjing University of Information Science and Technology for both my bachelor’s and master’s degrees. My work is the design and deployment of AI algorithms. Most of that experience is with large-language-model agents.

I do not write this book as a mathematician. I write it as an engineer who needs a way to keep up.

Why this book

The title says mathematics. I would not claim that much. What I can put on the page, for now, is a set of symbols, a few arguments, and the habit of asking what object a new method is actually using.

The first reason for the book is practical. Artificial intelligence changes quickly. A new name appears; the underlying idea is often an old one in another costume. If I can keep a private map of the theory—written in public, so that I cannot cheat—then the next frontier is easier to read, and easier to take back into my work.

The second reason is a conflict of interest. My job is agents and systems that ship. Product detail, proprietary stacks, and client work do not belong here. Theory is the part of the subject that least collides with that work. So the book stays on structures, not on implementations I cannot share.

What this book does not claim

This book claims no originality.

There is no new theorem, no new model, and no new method here that I present as mine. The arguments come from well-known and mainstream papers, textbooks, and monographs. Some views and explanations also come from public talks and videos on the internet. Where I can name a source, I will. Where a line of thought is common property of the field, I will say so rather than dress it as a discovery.

If a passage looks sharp, the sharpness belongs to the source, not to this manuscript. My work is arrangement, translation into one map, and the connections I need in order to keep reading.

Two threads

Two threads run through the text.

The first is mathematics—or, more honestly, the symbols and arguments I need. The assumed start is a first course in calculus from a university education. From there the book collects the ideas that keep returning: linear structure, probability, stochastic processes, convex optimisation, information, and dynamics. This is a working inventory, not a substitute for a specialist monograph in any one of those fields.

The second is AI. The book follows representations, learning rules, generative models, and the systems built on them, and fills in the mathematics at the moment it is needed. Where a derivation and a deployed system can be placed on the same page without pretending they are the same thing, they will be.

Once those two threads are in place, I want room for other fields—physics, finance, and whatever else shares a structure rather than a metaphor. Those subjects are later topics. They are not a second spine.

$$\text{Mathematics} \leftrightarrow \text{AI}, \quad \text{then other fields, when the bridge is real.}$$

How the book is organised

After this preface comes a notation list in four groups—calculus; matrices and vectors; probability, measure, and function spaces; integral transforms—plus a short list of objects used in later parts. Chapters will point back to those tables. The book then has three parts, and references at the end.

Part I — Mathematical Foundations

Part I lays out the language used later: spaces and inner products; maps and spectra; probability and expectation; information; optimisation; dynamics; and objects that are not flat vectors. Stochastic processes and convex optimisation belong here when they are needed, not as a complete course.

Part II — Mathematics of AI

Part II starts from the models. Representation, risk, generalisation, deep networks, and generative families are read as objects, then the missing mathematics is supplied, and—where it helps—tied to the kind of engineering that actually runs.

Part III — Intelligence and Beyond

Part III asks what those objects have to do with intelligence, and what can be borrowed from elsewhere without pretending the borrowing is a proof: reasoning, memory, planning, agents, and later bridges to physics, finance, and related fields.

How to read

The book need not be read in one straight line. You can walk from a piece of mathematics toward a model, or from a model back to the symbols that make it possible:

$$\text{Mathematics} \rightarrow \text{AI}, \quad \text{AI} \rightarrow \text{Mathematics}.$$

Definitions, theorems, corollaries, and examples share one chapter-wise counter (Definition 1.1, Theorem 1.2, ...). Proofs and remarks are unnumbered. Draft passages mark what is still open. Speculation is never presented as fact.

Background

A first course in calculus is the intended start. Linear algebra and elementary probability help, but they will be rebuilt when a later chapter depends on them. Programming may make some examples clearer. The centre of the book is still the argument, not the repository of code.

Symbols are defined at first use. The standing lists after this preface are grouped so that the text and the notation tables can refer to each other.

A living book

This is a working manuscript. It will change as the field changes, and as my own map of it changes. The source, two PDF editions, a website, animations, slides, and small examples live here:

<https://github.com/Martin-WMM/mathematics-of-intelligence>

The companions illustrate the text; they do not replace it. If the book does its job, a new paper will take less time to place—and the placement will be honest enough to use at work.

Notation

Symbols are defined at first use. The tables below are the standing vocabulary, grouped so that a later chapter can point back here—and so that a symbol here can point forward to the chapter that uses it.

This list will grow. It is not a course in any of the subjects named.

Calculus

The intended start is a first university course in calculus. Table N.1 is the core; derivatives and integrals return in Chapters 5 and 6.

Table N.1: Calculus. Maps, derivatives, integrals, and asymptotic notation. Later chapters point here when a rate, a gradient, or an integral is the object in play.

Symbol	Meaning
$\mathbb{N}, \mathbb{Z}, \mathbb{R}, \mathbb{C}$	Natural, integer, real, and complex numbers
$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$	A map from an n -vector to an m -vector
$\partial_i f, \nabla f$	Partial derivative; gradient
$\nabla^2 f, Hf$	Hessian of a scalar f
Df, J_f	Derivative / Jacobian of a vector-valued map
$\frac{d}{dt}$	Ordinary derivative in a scalar parameter
$\int_{\Omega} f$	Integral of f over a domain Ω
$o(\cdot), O(\cdot)$	Little-o and big-O asymptotics
$\arg \min, \arg \max$	Minimiser and maximiser of an objective

Matrices and vectors

Inner products and norms are the language of Chapter 1. Spectra and decompositions are taken up in Chapter 2.

Probability, measure, and function spaces

Expectation and conditionals are fixed in Chapter 3. Information quantities return in Chapter 4. Hilbert-space language is used whenever a score is an inner product, including later generative chapters.

Table N.2: Matrices and vectors. Linear algebra used as a language for representation, scores, and parameter updates. The geometry of $\langle u, v \rangle$ is Figure 1.1 in Chapter 1.

Symbol	Meaning
$x \in \mathbb{R}^d$	A real vector of dimension d
$x^\top, x_i$	Transpose; i -th coordinate
$\langle u, v \rangle$	Inner product
$\ x\ , \ x\ _p$	Induced norm; ℓ^p norm
$A \in \mathbb{R}^{m \times n}$	A real matrix with m rows and n columns
$A^\top, A^{-1}, A^+$	Transpose, inverse, and Moore–Penrose pseudoinverse
$\text{tr} A, \det A, \text{rank } A$	Trace, determinant, and rank
$I_d, \text{diag}(v)$	Identity of size d ; diagonal matrix from a vector
$\ A\ _2, \ A\ _F$	Operator (spectral) norm; Frobenius norm
$\lambda(A), \sigma(A)$	Eigenvalues; singular values
$uv^\top, A \otimes B$	Outer product; Kronecker product

Table N.3: Probability, measure, and function spaces. Random objects, measures, and L^p / Hilbert spaces. Measure-theoretic and functional-analytic names appear only when a later argument needs them.

Symbol	Meaning
$(\Omega, \mathcal{F}, \mathbb{P})$	A probability space
$X \sim p$	A random variable with law p
$p(x), p(x \mid y)$	Density or mass; conditional
$\mathbb{E}[X], \text{Var}(X), \text{Cov}(X, Y)$	Expectation, variance, and covariance
$\mathbf{1}_A$	Indicator of a set A
δ_x	Dirac measure at x
$\mu, \frac{d\mu}{d\nu}$	A measure; Radon–Nikodym derivative
$L^p(\mu)$	Functions with integrable p -th power
$\mathcal{H}$	A Hilbert space
$\langle \cdot, \cdot \rangle_{\mathcal{H}}$	Inner product on $\mathcal{H}$
T^*	Adjoint of a linear operator T
$\text{KL}(p \parallel q)$	Kullback–Leibler divergence

Integral transforms

Transforms turn a function on one domain into a function on another. They will be invoked from dynamics, scores, and generative models—in particular Chapter 6 and Chapter 16—rather than treated as a separate part.

Table N.4: Integral transforms. Fourier, Laplace, convolution, and the characteristic function. Each is a change of coordinates, not a new kind of object.

Symbol	Meaning
$\mathcal{F}f, \hat{f}$	Fourier transform of f
$\mathcal{F}^{-1}\hat{f}$	Inverse Fourier transform
$\text{Lap}\{f\}$	Laplace transform of f
$(f * g)(x)$	Convolution of f and g
$\varphi_X(t)$	Characteristic function $\mathbb{E}[e^{itX}]$
$M_X(t)$	Moment-generating function $\mathbb{E}[e^{tX}]$, when it exists

Objects in later parts

These names are not mathematics in themselves. They label the things Part II and Part III manipulate.

Table N.5: Objects in later parts. Loss, data, and model class as they appear from Part II onward. They sit on top of the tables above, not beside them.

Symbol	Meaning
$\mathcal{L}$	Loss or objective
$\mathcal{D}$	Data set or empirical measure
$\mathcal{M}$	Model class

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Part I

Mathematical Foundations

Before a model can be written down, the objects it acts on must have names: spaces, maps, measures, information, objectives, and dynamics. This part builds that language. Artificial intelligence appears only as motivation, not as the object of study.

Chapter 1

Spaces, Vectors, and Inner Products

A vector $x \in \mathbb{R}^d$ is not merely a container. Its dimension bounds capacity; an inner product defines similarity; a basis decides which variations count as independent. The geometry of that space is already a theory of what can be compared.

Raw observations—pixels, tokens, sensor readings—become usable only after they are placed in such a space. Figure 1.1 is the geometry that later chapters reuse as a score.

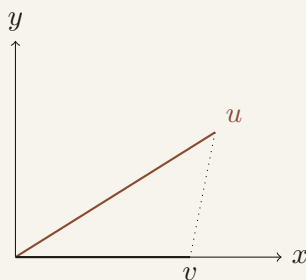


Figure 1.1: Inner product in the plane. The vector u and a reference direction v . The dotted segment is the drop that makes $\langle u, v \rangle$ visible as a signed length along v .

♦ **Definition 1.1** (Inner product). An *inner product* on a real vector space V is a bilinear map $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$ that is symmetric and positive definite: $\langle x, x \rangle \geq 0$, with equality if and only if $x = 0$. The induced norm is $\|x\| = \sqrt{\langle x, x \rangle}$.

■ **Theorem 1.2** (Cauchy–Schwarz). For all $u, v \in V$,

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Proof. If $v = 0$ the claim is immediate. Otherwise set $t = \langle u, v \rangle / \|v\|^2$ and expand the inequality $\langle u - tv, u - tv \rangle \geq 0$. ■

□ **Corollary 1.3** (Cosine bound). If u and v are nonzero, the cosine of the angle between

□ *them satisfies $|\cos \theta| \leq 1$, where*

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}.$$

- **Example 1.4** (Cosine similarity). Token or image embeddings are compared by the same cosine. A score near 1 means the two vectors point the same way—not that they are close in Euclidean distance.

Remark. Later chapters return to Definition 1.1 when attention, retrieval, and contrastive learning reuse inner products as scores.

Chapter 2

Linear Maps, Spectra, and Decompositions

A linear map A sends structure in one space to structure in another. Eigenvalues, singular values, and the associated decompositions name the independent modes of that action.

This chapter develops the spectral language used later for principal directions, graph operators, attention, and the forward process of a diffusion.

Chapter 3

Probability and Expectation

Risk, likelihood, and sampling are statements about a probability space. Expectation $\mathbb{E}$ turns a random object into a scalar; concentration says how far a sample can drift from that scalar.

The chapter fixes notation for random variables, conditionals, and empirical measures before learning and generation treat those objects as data.

Chapter 4

Information and Compression

A representation trades fidelity for efficiency. Information theory names that trade: entropy, mutual information, and rate–distortion say what can be kept when a channel or a latent code is finite.

Compression is not only an engineering constraint. It is a hypothesis about which bits of the world matter for a downstream task.

Chapter 5

Optimization as Geometry

Gradient descent is a discrete dynamical system on a loss landscape. Convexity makes local information global; without it, we only have local sensitivity.

Matrix calculus tells us how a scalar loss depends on high-dimensional parameters. The algorithm that follows from that derivative is not unique—the geometry of the parameter space still has to be chosen.

Chapter 6

Dynamics and Iteration

Each update is a map $\theta_{t+1} = T(\theta_t, \xi_t)$ driven by stochastic increments. Stability, noise, and step-size schedules are properties of that dynamical system.

The same language later describes learning algorithms, denoising flows, and closed-loop control. The chapter treats iteration as an object, not as a decoration on “running an optimiser”.

Chapter 7

Structure Beyond Vectors

Not every object is a flat list of coordinates. Sequences, graphs, fields, and manifolds carry relations that a naive vectorisation destroys.

When the object itself has structure, the representation should respect it: permutation symmetry on graphs, locality on images, temporal order on language. The mathematical question is which invariants to preserve and which to discard.

Part II

Mathematics of AI

With the language of Part I in hand, this part treats modern artificial intelligence as a collection of mathematical objects: representations, risk functionals, generalisation, deep parameterisations, and generative models.

Chapter 8

Representation

Contemporary models learn the encoding instead of hand-crafting it. Embeddings, autoencoders, and contrastive objectives all choose a space so that later linear operations become meaningful.

Multimodal systems go further: they align images, text, and actions in a shared space. Whether that space is universal or always task-relative remains an open question of this part.

Chapter 9

Learning, Loss, and Risk

Desired behaviour is reduced to a scalar. Population risk $\mathbb{E}[\ell(f(X), Y)]$ is the quantity we care about; empirical risk on a finite sample $\mathcal{D}$ is what we can compute.

The gap between the two is the first statistical problem of learning. Assumptions about the data-generating process must stay explicit.

Chapter 10

Generalization

Training loss is not the goal. Capacity, explicit regularisers, and the implicit bias of the optimiser jointly decide which interpolating solution is preferred.

Classical bounds and modern overparameterised practice do not yet share one theorem. This chapter keeps that tension visible.

Chapter 11

From Classical Learning to Deep Learning

Feature engineering and parameter estimation used to be separate. Depth couples them: the representation is learned by the same optimisation that fits the predictor.

Backpropagation is the computational consequence of that coupling. The conceptual shift is that representation and learning are no longer sequential stages.

Chapter 12

Modeling Distributions

A generative model states what “plausible” means. The statement may be a normalised density, an unnormalised energy, or a procedure that only yields samples.

Maximum likelihood is one way to fit such a model. It is not the only way, and it is not always tractable.

Chapter 13

Autoregressive Factorization

A joint distribution over a sequence factorises as a chain of conditionals. Language models are the canonical case: each token is generated given its past.

The mathematical object is a product of conditionals. The engineering object is a Transformer that parameterises those conditionals.

Chapter 14

Latent Variable Models

When the explicit density is intractable, we introduce latent variables and optimise a bound—typically an ELBO—via amortised inference.

The variational family is a modelling choice with the same status as the decoder. It should not be treated as a mere implementation detail.

Chapter 15

Adversarial and Implicit Generation

Some generators never write down a density. They learn a map from noise to data by playing a game against a discriminator.

The formulation is game-theoretic. Mode collapse, equilibrium concepts, and the absence of a likelihood are part of the mathematics, not just training anecdotes.

Chapter 16

Diffusion, Scores, and Flows

Iterative generation corrupts data with noise and learns to reverse the corruption. Scores, SDEs, and deterministic flows are different coordinates on the same idea: transport a simple distribution onto a data distribution.

The conceptual graph is more important than the product name:

noise $\rightarrow$ score $\rightarrow$ SDE/ODE $\rightarrow$ sampler.

Part III

Intelligence and Beyond

Representation, learning, and generation are not yet agency. This part asks how prediction becomes reasoning, how memory and planning enter, and what remains open once those structures are in use.

Chapter 17

Prediction versus Reasoning

Completion of a pattern is not the same as inference under constraints. Language models blur the distinction because fluent continuation often looks like reasoning.

The chapter separates correlation, search, and proof. It does not treat chain-of-thought traces as evidence that the underlying object is a proof system.

Chapter 18

Memory and Retrieval

Parametric memory lives in weights. External memory lives in an index. Both retrieve by geometry: nearest neighbours in a representation space, or attention over a context window.

Limits of context and limits of indices are different failure modes. They should not be collapsed into one slogan about “RAG”.

Chapter 19

Planning and Control

Actions, rewards, and value functions turn prediction into decision. Planning is constrained search in a model of the future; control theory studies the same loop with different notation.

The shared structure is a decision process. The book uses that structure rather than a catalogue of algorithm names.

Chapter 20

World Models

A world model is an internal simulator: a latent state, a transition, and a prediction used in service of action.

Whether the simulator must be physically faithful, or only useful for control, is an empirical and mathematical question—not a slogan.

Chapter 21

Agents

An agent couples perception, memory, planning, and action in an environment, often with tools and other agents.

Game-theoretic equilibrium, POMDPs, and language-conditioned policies are candidate structures. This chapter does not close the definition of intelligence; it keeps the habit of asking what structure is actually in use.

Chapter 22

Beyond

The first two parts give a language and a laboratory. This chapter does not close a definition of intelligence. It records what remains open, and which other sciences may enter once a shared structure is genuine.

Finance, physics, and control appear here only as candidate bridges—never as a second spine.

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